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Research article

The η-Al_5Fe_2 phase in the Al–Fe system: The issue with the sublattice model

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ABSTRACT

The thermodynamic descriptions of a non-stoichiometric crystalline phase used within the framework of the CALPHAD approach has to include an appropriate model to estimate the configurational entropy, e.g. within the compound energy formalism. Thereby, typically a sublattice model based on the known crystal structure data of the considered phase is employed. In the present work a revised thermodynamic description of the Al–Fe system is presented with focus on the sublattice model for the η-Al_5Fe_2 phase. In view of its channel structure containing Al+Fe as channel atoms with largely composition-independent atomic density (and not Al atoms and vacancies as believed previously), a new sublattice model $Al_4Fe_2(Fe,Al)_m$ with m = 1.5 is proposed. In contrast to previously proposed sublattice models, the new model is well able to reproduce the experimentally observed homogeneity range of the η-Al_5Fe_2 phase in the course of thermodynamic modelling. Thereby, in order to derive the new model, it was necessary to transfer the crystallographic model used during structure refinement specifically to a sublattice model for the thermodynamic modelling, in order to naturally represent the otherwise easily overestimated the configurational entropy.

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1. Introduction

Research on the thermodynamics of the binary Al–Fe system is driven, in particular, by the relevance of alloys rich in iron and aluminum in different materials applications. The Fe-rich side of the system is of interest in view of the application of Al as alloying element in steel [1–4] and activities to develop novel types of *iron aluminide* high-temperature alloys for structural applications [5] or as protecting coatings [6]. Information on the Al-rich intermetallic phases in the Al–Fe system is relevant as also these are considered as coatings [7,8]. Such intermetallic phases, furthermore, are product of the reactive interdiffusion in all kind of joints between Al- and Fe-base alloys [9], and as more or less detrimental particles in secondary Al and Al–Si alloys [10–13].

In order to capture the thermodynamics of the Al–Fe system, a series of thermodynamic descriptions have been formulated within the CALPHAD (CALculation of PHAse Diagrams) approach [14–24]. Earlier works, in particular, focused on the details of the ordering behavior of the Fe-rich bcc-based phases (bcc: body centered cubic) and treated some high-Al content phases (ζ-Al_2Fe, η-Al_5Fe_2 and θ-$Al_{13}Fe_4$) as stoichiometric. More recent works [22,23] shifted their attention to the Al-rich region with the afore-mentioned, structurally relatively complex intermetallic phases ζ, η and θ as well as the ε-Al_8Fe_5 high-temperature phase, considering newly available data on their homogeneity ranges [22,25,26] and on their heat capacities [27,28].

In the course of CALPHAD-type thermodynamic modelling of crystalline phases, the homogeneity range of a particular phase is typically described within the so-called compound-energy formalism (CEF) [29]. Thereby, a sublattice model with each sublattice being occupied by one or several different species and where ideal configurational entropy is assumed to occur due to mixing on each of the sublattices (see Appendix A for details). The ideal configurational entropy is, however, a sometimes severe approximation [30], which is not easily overcome by computational methods which can be easily implemented into standard CALPHAD descriptions suitable also for multicomponent systems. In any case, selection of a sublattice model consistent with the real crystal structure should help at arriving at a "good" thermodynamic description of experimental input data by the thermodynamic model. Even more, physically meaningful modelled parameters and sublattice occupancies can be achieved which can be compared with real crystallographic data.

One of us (OF) has coauthored one of the above-referenced thermodynamic descriptions [22] considering sublattice models for

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various phases, which were derived from the structure models of previous crystallographic works. The homogeneity ranges predicted by that description for the η-Al_5Fe_2 phase (but also, to some extent, for the ζ-Al_2Fe phase), however, strongly differ from experimental data. Some independent issues with re-stabilization of the ζ and θ phases above 2400 K were addressed in a subsequently published PhD thesis [24] by introducing high-temperature ranges for the parameters describing the Gibbs energies of end-members of the respective phases. However, this was not implemented properly, resulting in re-stabilization of ζ-Al_2Fe above 4500 K and caused large jumps in enthalpy and entropy for the ζ and θ phases at the temperature interval boundary (for example, about 270 kJ·mol^{-1} at 2500 K in the case of ζ-Al_2Fe). It should be noted that the issues with meeting the homogeneity ranges were absent in a later work [23], which, however, used models without recourse to the actual crystallography of these two phases.

The works on Ref. [22,24] had just been finalized when new insight into the crystal structure of the disordered η-Al_5Fe_2 phase emerged [31], with results revising, to some extent, earlier works [32,33]. For ordered low-temperature phases of η-Al_5Fe_2, see [31,34–37]. The present work makes use of this new insight by revising the sublattice model η-Al_5Fe_2. Moreover, some corrections were done on the ζ and θ phases (due to the issues mentioned above), as well as on the ε-Al_8Fe_5 high-temperature phase (due to some unreasonably large temperature dependence of some parameters from [22]). Doing so, not all issues with respect to the thermodynamic description of the Al–Fe system are resolved. Nevertheless, it is shown that the new sublattice model for the η-Al_5Fe_2 phase allows for a much superior thermodynamic modeling of this phase than was possible previously [22].

2. Measures for thermodynamic modelling

2.1. Sublattice model for the η-Al_5Fe_2 phase

Fig. 1 illustrates the developing state of knowledge on the atomic structure of the η-Al_5Fe_2 phase. The crystal structure of the η-Al_5Fe_2 phase has, for the first time, briefly been reported by Schubert et al. [33] as having orthorhombic *Cmcm* symmetry based on single-crystal diffraction analysis. 8 Al(1) and 4 Fe atoms were located within this unit cell, fully occupying Wyckoff sites having the corresponding multiplicities. Moreover, a further 4-fold position was reported to be partially occupied by Al (Al(2) sites; estimated occupancy 0.7), arriving at an estimated average unit cell content of $Al_{8+4\times0.7}Fe_4$ (=2 × $Al_{5.4}Fe_2$). Burkhardt et al. [32] used more advanced single-crystal based diffraction data and refinement methods, to arrive at structure parameters of the η-Al_5Fe_2 phase, which were also discussed in some depth. In the course of a more detailed structure discussion, the authors identified an $Al(1)_4Fe_2$ framework structure (Fig. 1a) hosting, per unit cell, two channels running along the [001] direction (see Fig. 1b), being equivalent by the *C*-centering translation of the space group symmetry. During structure refinement the channels appear to contain residual electron density, which had led Schubert et al. [33] to introduce the partially occupied Al(2) sites (see Fig. 1c). Burkhardt et al. [32], however, found the electron density to be more continuous. They obtained a better fit to the observed diffraction intensity data by additionally introducing Al(3) sites, also with refined partial occupancy by Al (see Fig. 1d). Refined atomic displacement parameters further reveal that the electron density is additionally spread out into the [001] direction for both the Al(2) and Al(3) sites. Based on the refined occupancies of the Al sites in the channels (Al(2): 0.32; Al(3): 0.24), Burkhardt et al. [32] arrived at a unit cell content of $Al_{8+4\times0.32+8\times0.24}Fe_4$ (=2 × $Al_{5.6}Fe_2$), and thus at a higher overall Al content in the channels than estimated by Schubert et al. [33]: 1.6 instead of 1.4 Al atoms per channel and length of c_η (see Fig. 1). Notably, this Al content was significantly higher than implied by the nominal Al_5Fe_2 composition of the single-phase alloy, and the authors already speculated on the possible presence of Fe in the channels.

In the course of their thermodynamic modelling, Zienert & Fabrichnaya [22] started by considering a sublattice model derived from the structure model by Burkhardt et al. [32]. In view of this, the fully occupied 8 Al(1) and 4 Fe sites and the partially occupied Al(2) and Al(3) sites with 4- and 8-fold multiplicity lead to a sublattice model for the compound energy formalism of $Al_4Fe_2(Al,Va)_4(Al,Va)_2$ (Va for vacancies; normalizing here and in the following the formula

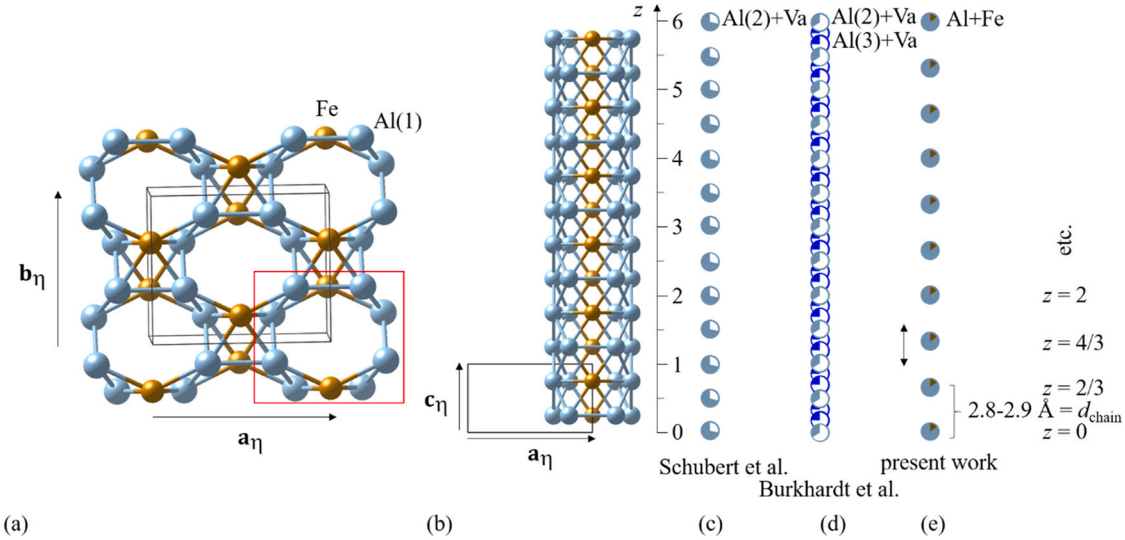

Fig. 1. Developing insight into the crystal structure of the disordered η-Al_5Fe_2 phase with the black box depicting the orthorhombic unit cell adopting the choice of origin due to Schubert et al. [33]. (a) Al(1) and Fe atoms forming a framework structure containing channels, with one channel being highlighted by a red rectangle. (b) Channel framework and proposed channel filling (c-e), with the sphere filling depicting the partial occupation by Al+Va or Al+Fe. (c) Channel filling due to Schubert et al. [33] and (d) Burkhardt et al. [32] derived from single-crystal diffraction data, assuming partial occupation of two (types of) channel sites, Al(2) and Al(3). (e) Channel filling with a largely equidistant chain of Al+Fe atoms concluded by Becker & Leineweber [31] from the structures of the low-temperature phases and the average electron density in the high-temperature phase. These insights led to formulation of the *New* model in the present work. From the average atomic distance d_{chain} this chain is incommensurate with c_η (see text), while here it was approximated to be have an atomic distance of $2/3c_\eta$. Thereby, the double-arrow indicates that no fixed origin of the chain along the [001] direction is assumed.

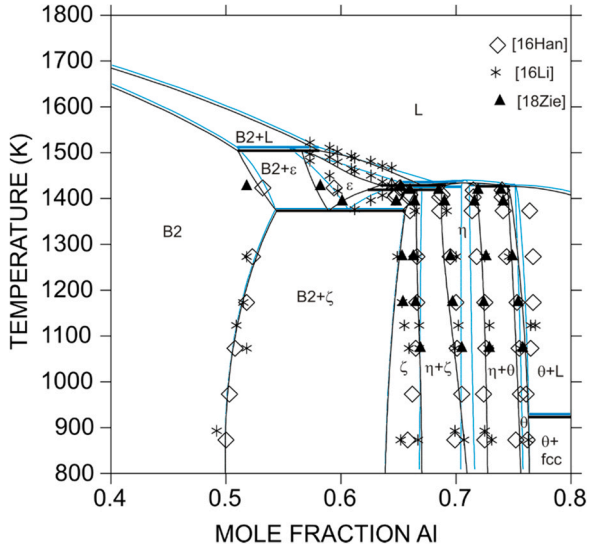

Fig. 2. Part of the Al–Fe phase diagram calculated based on the database from Zienert & Fabrichnaya [22] (blue lines), exhibiting limited agreement with the experimentally determined phase boundaries as it concerns the η-Al_5Fe_2 phase (black data points; [22,25,26]). The black lines represent the phase diagram calculated based on the presently derived database using the modified thermodynamic model for the η phase (*New*) based on the channel occupation as illustrated in Fig. 1(b). A much better agreement with the experimental data points delimiting the homogeneity range of the η phase is encountered while a considerable disagreement remains with respect to the ζ phase.

to the contents of half of the orthorhombic unit cell). In order to reduce the number of parameters during the thermodynamic modelling, the sublattices formed by the two types of channel sites, Al(2) and Al(3), were merged to obtain the model called *Zienert1*:

Zienert1: $Al_4Fe_2(Al,Va)_6$.

As, however, indicated in [22], quite unconvincing results were obtained upon thermodynamic modelling using this model. Hence, the model was modified towards:

Zienert2: $Al_4Fe_2(Al,Va)_2$:

narrowing also down the existence range of the solution model from 66.7 at. %–83.3 to 66.7 at. %–75 at. % of Al. The number of channel sites in the *Zienert2* model corresponds to the number considered by Schubert et al. [33]. Although the parameters due to do this model could be optimized, the resulting fit of the homogeneity range of the η-Al_5Fe_2 phase to the experimental data was not fully convincing (see blue phase boundaries in Fig. 2).

By the time when the thermodynamic description of the Al–Fe system by Zienert & Fabrichnaya [22] got finalized, new crystallographic insight was obtained on η-Al_5Fe_2. This insight concerned the channel occupation in the high-temperature phase η-Al_5Fe_2 [31] but also on ordering of the atoms in at least four different low-temperature phases [31,34–38]. Evaluation of powder X-ray diffraction data from different alloy powders of the disordered η-Al_5Fe_2 phase having Al contents in the range of 70.6–73.0 at. % gave intriguing results: emanating from the presence of only Al on the channel sites in view of [32,33], the overall channel occupancy by Al refined by the Rietveld method appeared to decrease with increasing Al content of the alloy/in the η-Al_5Fe_2 phase. Hence, an alternative structure model was proposed and confirmed by Rietveld refinements. The confirming Rietveld refinements were performed based on diffraction data of η-Al_5Fe_2 of various Al contents. The structure model for refinement was further developed based on the structure model by Burkhardt et al. [32]. It was modified by allowing partial occupation of the Al(2) and Al(3) sites (see Fig. 1(d)) by Al, Fe and vacancies while the overall unit cell content was constrained to comply with the respective composition of the η phase. The refined total atomic content corresponds to about 1.47–1.5 Al+Fe atoms per channel and length c, independent of the composition of the η-Al_5Fe_2 phase. Such a model implies an average atomic distance in the order of d_{chain} = 2.8–2.9 Å within the channels, a value reasonably agreeing with the shortest Al(1)–Al(1) distances within the framework structure (see Fig. 1(a)). Hence, the channels can be conceived as filled by atomic chains Al+Fe atoms with approximately constant distance without a large number of missing atoms (vacancies; see Fig. 1(e)). Thereby, this distance d_{chain} is incommensurate with respect with the periodicity framework structure, i.e. d_{chain}/c_η is irrational so that there is no distinct origin of the atomic chain in the channels with respect to the framework structure, if a part of the structure is depicted. This fact is emphasized by the double arrow in Fig. 1(e): the depicted part of the atomic chain can be shifted as a whole along whatever distance to yield a valid occupation pattern for the chain.

Long-range order with respect to the *relative* shifts of these chains appears to be absent in the sufficiently rapidly quenched η-Al_5Fe_2 phase, as also implied by layers of diffuse scattering perpendicular to the [001] direction (not reported in Ref. [31] but shown briefly in Ref. [39]). In the low-temperature phases, long-range order exists with respects to the shifts of the atomic chains within the different channels, as worked out for the low-temperature phases η′ [34,35], η′′ and η′′′ [31,36].[1] The structure of η′ has a significantly reduced occupation of the channel sites by only 1.33 atoms per channel (likely only Al; Al_8Fe_3) and length c_η, so that the (apparently pure Al) chains become commensurate with the framework structure. In the two incommensurate composite structures η′′ and η′′′, the values of d_{chain} are directly obtained as length the c-axis of the second composite part of the composite structure model (a_{23} in Ref. [31]). Assuming one atom per unit cell of the second composite part, there are 1.42–1.48 (= c_η/d_{chain}) atoms per channel and length c_η which is close to the value of about 1.47–1.5 assessed for the high-temperature η phase. Note that additionally to the positional order in the channels the works do not reveal any long-range order of Al against Fe on the channel sites, although it has been suggested based on single-crystal X-ray diffraction data that the Fe atoms might be enriched on Al(3)-like positions within the channels [36].

It is now proposed that the described type of occupation of the channels by Al and Fe atoms might imply the use of a sublattice model of the type.

New: $Al_4Fe_2(Fe,Al)_m$.

with $m = c_\eta/d_{chain}$ being slightly smaller than 1.5 and not necessarily being rational. By doing so, only the assumedly random Al +Fe mixing contributes to configurational entropy, but not the disordered displacements of the chains of atoms in different channels with respect to each other. For simplicity we have adopted m = 1.5, whereby the chains in the channels would get commensurate with c_η (but has no special consequences for the thermodynamics), i.e. the atomic distance becomes $d_{chain} = 2/3c_\eta$. The reader is reminded at the double arrow in Fig. 1(e), negating any explicit positioning of the chain of atoms with respect to the unit cell origin is made. It is simply assumed that each channel is filled with a chain of atoms having some regular distance d_{chain} implying the value of m and ideal occupational mixing of Fe and Al on these sites.

2.2. Further measures upon modelling

The ε-Al_8Fe_5 high-temperature phase exhibits an I-type γ-brass structure. Stein et al. [40,41] used high-temperature neutron diffraction data on some Fe-rich composition (< 61.5 at. % Al, corresponding to Al_8Fe_5) to reveal an atomic distribution corresponding to $Al_2Fe_2Fe_3(\mathbf{Al},Fe)_6$ [40].[2] It is, however, unknown how Al-rich

[1] There is even evidence for the fourth long-range ordered phase η^m without a worked out structure model [37,38].

compositions are realized. In view of the situation experimentally known from Cu_5Zn_8 (relying on similarity with the defect characteristics of Cu_5Zn_8) [42], Zienert & Fabrichnaya [22] proposed a sublattice model $Al_2Fe_2(Al,\mathbf{Fe})_3(\mathbf{Al},Fe)_6$. For simplification, the two mixed-occupancy sublattices were merged towards $Al_2Fe_2(Al,Fe)_9$. This, however, might appear inadequate in retrospective because the third and fourth sublattices have opposite preferential occupancies. In the present work, instead of going back to an unmerged sublattice model, a sublattice model $(\mathbf{Al},Fe)_2Fe_2(Al,\mathbf{Fe})_3Al_6$ as previously used by Rank et al. [23] for the ε-Al_8Fe_5 phase and as introduced earlier by Witusiewicz et al. [43] for the chemically similar phase in the Al–Cr system is applied.

For the ζ-Al_2Fe phase (sublattice model $Al_{12}Fe_5(Al,Fe)_2$ in view of [44]) and for the θ-$Al_{13}Fe_4$ phase (sublattice model $Al_{0.6275}Fe_{0.235}(Al,Va)_{0.1375}$; as already used in Ref. [14]) the same sublattice models were taken as in Ref. [22]. However, in order to avoid the issue of re-stabilization of these phases at high temperatures, which arises both in view of the databased presented in Ref. [22] and in Ref. [24], the increase in heat capacity above the melting temperatures of the ζ and θ phases was restrained by introducing a second temperature interval of the Gibbs energy description based on different temperature dependence of heat capacity. A similar approach was recommended by SGTE (Scientific Group Thermodata Europe) and used in the work of Dinsdale [45]. In the case of the ζ-Al_2Fe phase, a temperature-independent contribution of heat capacity to the Gibbs energy of the $Al_{12}Fe_5(Fe)_2$ end-member was introduced at temperatures above 2000 K. Thus, at 2000 K, the calculated heat capacity, enthalpy, entropy, and Gibbs energy are the same when using the equations for the low- and high-temperature intervals. A similar approach was implemented in the case of the θ-$Al_{13}Fe_4$ phase. It should, however, be noted that, as compared to [22], the temperature-dependent contributions to the Gibbs energies of both end-members of the θ phase were slightly modified for the low-temperature ranges. The resulting changes in the calculated phase diagram (see Fig. 2) were within the uncertainty ranges plot. For the A2 solid solution (and of its B2 ordered state) and for the liquid model, the parameter values were taken as the same as in Ref. [22].

Hence in the present work, only the parameters pertaining to the η, ε, and θ were reassessed using the same experimental data as considered in Ref. [22], by using the PARROT module of the Thermo-Calc software [46].

3. Results and discussion

The optimized parameters of the η, ε, ζ, and θ phases are presented in Table 1. The complete thermodynamic database is available as supplementary material. It is worth to note that the Gibbs energies of the end-members describing the η-Al_5Fe_2 phase are adjusted based on the heat capacity data from Refs. [22,27,28]. However, heat capacity data above 1200 K have not been taken into account due to their drastic increase just before melting of the η-Al_5Fe_2 compound (1434 K) in accordance with recommendation of Lukas et al. [47]. The calculated temperatures and phase compositions for invariant reactions are compared with experimental data in Table 2, demonstrating reasonable agreement. Moreover, the thermodynamic quantities for the η, ζ, and θ phases are compared with data from [22,27,48] in Table 3.

The phase diagram of Al–Fe calculated using the new thermodynamic description together with experimental data is presented in Fig. 2. As visible by comparison with the calculated phase diagram due to [22] shown in blue in Fig. 2, the experimentally determined homogeneity range of the η phase is reproduced much better in the present than in the previous description [22]. Similar as in [22], the experimentally determined homogeneity range of the ζ phase is not properly modelled on the Fe-rich side. The shape of the homogeneity range of the θ phase appears to be slightly better met by the current description. As it concerns the ε phase, the poorer agreement with the homogeneity range data as compared to [22] appears acceptable in view of the unreasonably large temperature dependence of some parameters employed in its description.

It is proposed that the superior behavior of the sublattice model *New* (as compared to *Zienert1* and *Zienert2*) for the η-Al_5Fe_2 phase in getting a good fit to the experimentally observed homogeneity range is caused by a better prediction of the configurational entropy by that model. Fig. 4 depicts the calculated configurational entropy for the *Zienert1*, *Zienert2* and *New* models. As visible, the models *Zienert1* and *Zienert2* generate a much larger configurational entropy than it is the case for the *New* model. The origin of this large configurational entropy is actually primarily not the assumption that Al and vacancies instead of Al and Fe atoms occupy the channel sites. It is actually caused by the idea that every atomic site considered in the course of the structure refinement constitutes sites with a multiplicity that can directly be adopted for a sublattice scheme suitable for the compound energy formalism (CEF) with an assumedly ideal configurational entropy.

That a correspondingly calculated configurational entropy is unrealistically large becomes clear upon inspecting e.g. the Burkhardt [32] model for the distribution of sites in a channel, which has also been used by Becker & Leineweber [31] to refine the Al+Fe content in the channels, and which has been the basis for the *Zienert1* model. Consider e.g. at the fractional coordinate $z = 1$ (an Al(2) site; see Fig. 1(d) and in particular Fig. 3) and assume its occupation by an atom (Al or Fe). Then the next channel sites in the positive and negative [001] directions occur at distances of 0.8 Å (Al(3)), 1.4 Å (Al(3)), 2.11 Å (Al(2)) and 2.8 Å (Al(3)) (see Fig. 3). In view of the shortest realized Fe-Al(1) and Al(1)–Al(1) distances in the framework structure, which are longer than 2.5 Å, it becomes evident that from the above list, the sites having a distance < 2.5 Å are blocked by the atom occupying the position $z = 1$, so that only the Al(3) site in the distance of 2.8 Å can be occupied. Such site blocking [30] causes considerable short-range order, i.e. the sites are not occupied independently from each other and thus configurational entropy will be overestimated by the common formulas for ideal configurational entropy as adopted for the calculations leading to Fig. 4. Note that this site blocking also concerns the adjacent Al(2) sites at $z = 1/2, 3/2$. Hence, also if only the sites due to Schubert et al. [33] (see Fig. 1(a)) are considered for occupation, site blocking will cause an overestimated configurational entropy within the *Zienert2* model. Such site blocking effects are irrelevant in the *New* model which considers the really available numbers of sites for Al+Fe mixing.

Fig. 4 also includes an estimate for the evolution for the sublattice model considered by Rank et al. [23].[3]

Rank: $(Fe,\mathbf{Al})_5(\mathbf{Fe},Al)_2$,

whereby ideal preferential ordering was assumed for both sublattices, yielding zero configurational entropy for the Al_5Fe_2 composition. This *Rank* model has, however, been formulated without recourse to the crystallographic knowledge. It contains the apparent ideal composition Al_5Fe_2 at 71.4 at. % Al as one end-member, while

[2] The bold element symbols correspond to occupation in the lowest-energy configuration in the case of the stoichiometric composition. The sequence of sublattices in this and the following formulas for the I-type γ-brass structure corresponds to what is typically referred to as the inner tetrahedron (IT), outer tetrahedron (OT), octahedron (OH) and cuboctahedron (CO) shell structure frequently employed for structure description of such structures, e.g. [42].

[3] That model had already been used by Du et al. [18]. At the time of that work, however, the above-mentioned data on the homogeneity range of the η-Al_5Fe_2 phase were not available [22,25,26].

Table 1
Parameters of the thermodynamic database derived in the present work for the Al–Fe system in terms of numerical value equations yielding values in J/mol of the indicated formula unit while T is the temperature values in K. The parameters of the bcc based phases and of the liquid phase are the same as from [22], whereas the parameters for the second temperature ranges of the ζ and θ phases were adjusted in this work.

Phase	Model	Temperature, K	Parameter (J·mol^{-1})	Reference
η-Al_5Fe_2	$Al_4Fe_2(Al,Fe)_{1.5}$	298.15–3000	${}^0G^{Al_5Fe_2}_{Al:Fe:Al} = -234096.29 + 1004.3581\cdot T - 179.0366\cdot T\cdot \ln T - 0.02212\cdot T^2 + 964645.05\cdot T^{-1}$	This work
		298.15–3000	${}^0G^{Al_5Fe_2}_{Al:Fe:Fe} = -241978.841 + 999.6593\cdot T - 179.0366\cdot T\cdot \ln T - 0.02212\cdot T^2 + 964645.05\cdot T^{-1}$	This work
ε-Al_8Fe_5	$(Al,Fe)_2Fe_2(Al,Fe)_3Al_6$	298.15–6000	${}^0G^{Al_8Fe_5}_{Al:Fe:Al:Al} = +11\cdot GHSERAL + 2\cdot GHSERFE - 132424.021 - 27.088216\cdot T$	This work
		298.15–6000	${}^0G^{Al_8Fe_5}_{Fe:Fe:Al:Al} = +9\cdot GHSERAL + 4\cdot GHSERFE - 200677$	[23]
		298.15–6000	${}^0G^{Al_8Fe_5}_{Al:Fe:Fe:Al} = +8\cdot GHSERAL + 5\cdot GHSERFE - 255459.381 - 39.3071\cdot T$	This work
		298.15–6000	${}^0G^{Al_8Fe_5}_{Fe:Fe:Fe:Al} = +6\cdot GHSERAL + 7\cdot GHSERFE - 313712$	[23]
		298.15–6000	${}^0L^{Al_8Fe_5}_{Al,Fe:Fe:*:Al} = -13558.872 + 7.0041851\cdot T$[a]	This work
		298.15–6000	${}^0L^{Al_8Fe_5}_{*:Fe:Al,Fe:Al} = -57633.590 + 14.012992\cdot T$[a]	This work
		298.15–6000	${}^1L^{Al_8Fe_5}_{*:Fe:Al,Fe:Al} = -57639.363$[a]	This work
ζ-Al_2Fe	$Al_{12}Fe_5(Al,Fe)_2$	298.15–6000	${}^0G^{Al_2Fe}_{Al:Fe:Al} = -575167.8 + 2402.6807\cdot T - 432.1056\cdot T\cdot \ln T - 0.066401105\cdot T^2 + 1566274.5\cdot T^{-1}$	[22]
		298.15–2000	${}^0G^{Al_2Fe}_{Al:Fe:Fe} = -554383 + 893.2723\cdot T - 173.61174\cdot T\cdot \ln T - .6089367\cdot T^2 + 2.459436\cdot 10^{-4}\cdot T^3 - 5.311\cdot 10^{-8}\cdot T^4$	[22,24]
		2000–6000	${}^0G^{Al_2Fe}_{Al:Fe:Fe} = -2767770 + 13743.0612\cdot T - 1805.27\cdot T\cdot \ln T$	This work
θ-$Al_{13}Fe_4$	$Al_{0.6275}Fe_{0.235}(Al,Va)_{0.1375}$	298.15–6000	${}^0G^{Al_{13}Fe_4}_{Al:Fe:Al} = -31588.8807 + 144.73675\cdot T - 25.3674\cdot T\cdot \ln T - .001549605\cdot T^2 + 163227.5\cdot T^{-1}$	This work
		298.15–2500	${}^0G^{Al_{13}Fe_4}_{Al:Fe:Va} = -26391.8 + 89.607713\cdot T - 15.2734\cdot T\cdot \ln T - 0.0178165\cdot T^2 + 6.29955\cdot 10^{-6}\cdot T^3 - 1.27236\cdot 10^{-9}\cdot T^4$	This work
		2500–6000	${}^0G^{Al_{13}Fe_4}_{Al:Fe:Va} = -191336.4 + 845.783569\cdot T - 106.69\cdot T\cdot \ln T$	This work

[a] The parameters ${}^0L^{Al_8Fe_5}_{Al,Fe:Fe:*:Al}$, ${}^0L^{Al_8Fe_5}_{*:Fe:Al,Fe:Al}$, and ${}^1L^{Al_8Fe_5}_{*:Fe:Al,Fe:Al}$ are the same if sublattice is fully occupied by Al or Fe

Table 2
Temperatures (K) and phase compositions (molar fraction of Al) for the invariant phase equilibria in the Al–Fe system (L: Liquid) according to different databases from this work and the literature.

Reaction	This work	[22] calc/measured	[23]	[25]	[26,40]
L+B2⇄ε	1504	1505/1507	1505	1506	1504
	L:0.581	0.581	0.586	0.589	0.593
	B2:0.510	0.510	0.500	0.519	-
	ε:0.566	0.554	0.568	0.566	0.560
L⇄ε + η	1429	1429/1430	1429	1430	1428
	L:0.670	0.682	0.689	0.677	0.684
	ε:0.634	0.651	0.632	0.637	0.644
	η:0.691	0.706	0.695	0.683	0.699
L+ η⇄θ	1427	1423/1425	1424	1424	1423
	L:0.7470	0.750	0.748	0.743	0.766
	η:0.7173	0.712	0.714	0.714	0.715
	θ:0.7447	0.752	0.734	0.738	0.765
ε + η⇄ζ	1419	1419/1422	1420	1420	1418
	ε:0.623	0.645	0.63	0.641	0.642
	η:0.688	0.705	0.69	0.683	0.684
	ζ:0.666	0.671	0.66	0.654	0.668
ε⇄B2 + ζ	1373	1370/1369	1368	1368	1368
	ε:0.5893	0.606	0.589	0.607	0.609
	B2:0.5439	0.543	0.518	0.54	0.529
	ζ:0.6558	0.656	0.646	0.643	0.645
L⇄θ + fcc	923	923	926	928	927
	L:0.985	0.985	0.989	-	
	θ:0.763	0.764	0.764	0.765	
	fcc: 0.999	0.999	1.0	-	
L⇄η	1434	1433	1433	1433	1431
	0.706	0.708	0.7	0.7	

Table 3
Thermodynamic quantities (per mole atoms) for the η, ζ, and θ phases at 298.15 K.

Phase[1]	ΔH_m (kJ·mol^{-1})	$S^{298.15}$ (J·mol^{-1}·K^{-1})	ΔG_m (kJ·mol^{-1})	Reference
η-Al_5Fe_2	−30.5 ± 1.7			[47][2]
		23.00		[27][3]
	−23.09	29.97	−32.08	[22][4]
	−23.50	28.94	−32.01	This work[5]
ζ-Al_2Fe	−27.8 ± 1.9			[47][2]
		24.45		[27][3]
	−23.72	30.21	−32.73	[22][4]
	−23.80	29.96	−32.73	This work[5]
θ-$Al_{13}Fe_4$		23.65		[27][3]
	−21.77	29.04	−30.42	[22][4]
	−22.80	27.96	−31.14	This work[5]

[1] Stoichiometric composition is considered.
[2] Value determined by solution calorimetry.
[3] Estimated using heat capacity data.
[4] Calculated using the database where the *Zienert2* model is applied to describe the η-Al_5Fe_2 phase.
[5] Calculated using the database where the *New* model is applied to describe the η-Al_5Fe_2 phase.

with the other end-members the model is able to describe the thermodynamics for compositions in the range of 0–100 at. % Al. With appropriate model parameters a considerable degree of order exists at all relevant temperatures with antisite atoms on the respective sublattice realizing deviation from the ideal Al_5Fe_2 composition towards both Al- and Fe-rich compositions. It is expected that such a model will definitely be able to model with or without narrowing down the homogeneity range towards low temperatures, depending on the model parameters.

Accordingly, it is no surprise Rank et al. [23] have also been quite successful in describing the homogeneity range of the ζ-Al_2Fe phase using the (Fe,**Al**)$_2$(**Fe**,Al) sublattice model without recourse to the phase's actual crystal structure [44,49]. In fact, the poor reproduction of the homogeneity range achieved with the presently used sublattice model for the ζ phase derived from [44] (see Fig. 2 and

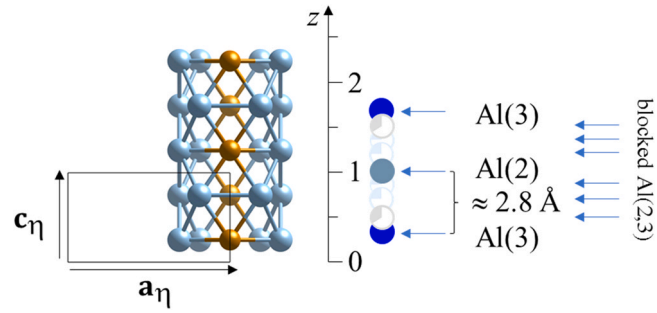

Fig. 3. Channel occupation with potential sites as considered by Burkhardt et al. [32] (compare Fig. 1(d)). It is assumed that initial Al atom is placed at $z = 1$ (Al(2), bright color). The adjacent channel sites get blocked by this atom (pale color). Consequently, the first of the indicated Al(2)/Al(3) sites, which can be occupied, are the indicated Al (3) atoms (bright colors) having a distance of about 2.8 Å corresponding approximately to the assessed d_{chain} in the model with mixed Al+Fe occupation.

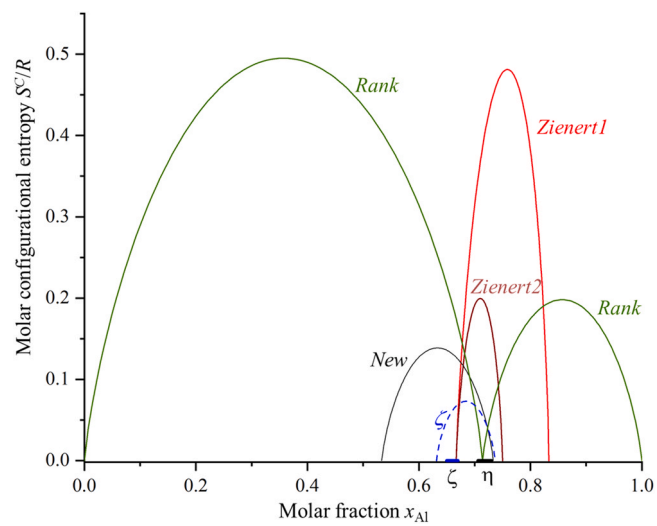

Fig. 4. Configurational entropy of η and ζ plot per mole of atoms obtained for different, indicated sublattice models. The fat lines on the abscissa illustrate the experimentally observed homogeneity ranges of these phases.

discussion above) might constitute another case where an incompletely known defect-structure model causes an inappropriate estimate for the course of the configurational entropy (see Fig. 4). It might be worth reconsidering the crystal structure of this ζ-Al_2Fe phase in some more detail, including the defect mechanisms which are responsible for realization of the experimentally observed homogeneity range.

The currently presented case of the η-Al_5Fe_2 phase may serve as an example, where a not-perfectly ordered crystal structure requires some special attention to set up an appropriate sublattice model for the compound energy formalism. Essentially, not every atomic model which yields an average electron density which well reproduces the intensities of the Bragg reflections is adequate for a 1:1 translation into a sublattice model for the compound energy formalism. Thereby, it must be kept in mind that most reported atomic structure models have been determined by crystallographers not having thermodynamic modelling in mind.

4. Conclusions

A revised thermodynamic description of the Al–Fe system within the framework of the CALPHAD method has been presented, with special emphasis on improving fit to the experimentally observed

homogeneity range of the η-Al_5Fe_2 phase. Thereby, the available crystallographic information on that phase has been reconsidered in detail to arrive at the new sublattice model which appropriately estimates the composition dependent configurational entropy due to the atoms located within channels of the crystal structure. It was shown that this new sublattice model should not be a 1:1 translation of the crystallographic description used during the previous structure refinements based on diffraction data. Such a model would lead to a strong overestimation of the configurational entropy due to site-blocking effects.

CRediT authorship contribution statement

M. Ilatovskaia: Methodology, Validation, Formal analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration **H. Becker**: Methodology, Data Curation, Writing - Review & Editing, Project administration **O. Fabrichnaya**: Conceptualization, Methodology, Formal analysis, Resources, Data Curation, Writing - Review & Editing, Supervision, Funding acquisition **A. Leineweber**: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, Funding acquisition.

Data Availability

Data will be made available upon reasonable request.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andreas Leineweber reports financial support was provided by German Research Foundation. Hanka Becker reports financial support was provided by German Research Foundation. M. Ilatovskaia reports was provided by German Research Foundation. Olga Fabrichnaya reports was provided by German Research Foundation.

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Appendix. A: Compound energy formalism and configurational entropy

For crystalline phases having several different crystallographic sites, which can be occupied by atoms or ions or more complex species, the compound energy formalism (CEF) developed by Hillert [29] can be applied to model the composition-dependent thermodynamics of these phases. Let’s assume a phase with two sublattices as represented by the formula $(A,B)_k(C,D)_m$, indicating that A and B species mix on the first sublattice while C and D mix on the second sublattice. The numbers k and m are stoichiometric indices for the first and second sublattice, respectively. The molar Gibbs energy G per formula unit for the phase is given as

$$G = \sum_i \sum_j y_i^{(1)} y_j^{(2)} G_{ij} - TS^C + G^E, \tag{1}$$

where $y_i^{(1)}$ and $y_j^{(2)}$ are the mole fraction of species i and j on the first and second sublattices, respectively; G_{ij} is the corresponding molar Gibbs energy of the end-member of the solution, in which the first sublattice is occupied only by the species i and the second only by the species j; T is the temperature; G^E is the excess Gibbs energy; and S^C is the configurational entropy which is defined as following

$$S^C = -R\left[n_1 \sum_i y_i^{(1)} \ln y_i^{(1)} + n_2 \sum_j y_j^{(2)} \ln y_j^{(2)} \right], \tag{2}$$

where n is the stoichiometric number of each sublattice and R is the gas constant.

The excess Gibbs energy is expressed using Redlich−Kister equation as

$$G^E = \sum_{(1)} y_i^{(1)} \sum_{(2)} y_j^{(2)} y_m^{(2)} L_{i:j,m}^{(2)} + \sum_{(2)} y_j^{(2)} \sum_{(1)} y_i^{(1)} y_k^{(1)} L_{i,k:j}^{(1)}, \tag{3}$$

where the parameter $L_{i,k:j}^{(1)}$ is related to interactions between species i and k on the sublattice (1) while $L_{i:j,m}^{(2)}$ is interaction parameter between species j and m on the sublattice (2). In the case of the present work, the only types of species are Al and Fe.

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2022.168361.

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